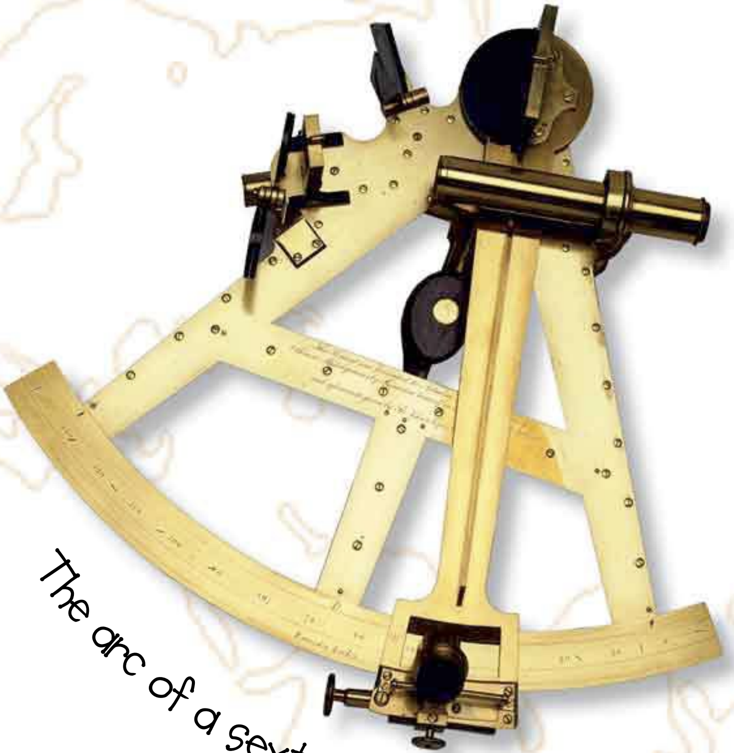
Mariner's astrolabe

Sailors used astrolabes, first made around 1300, to measure the **height of the Sun** or a particular star. This allowed them to calculate their ***latitude*** (north–south position). Mariner's astrolabes helped sailors **EXPLORE FARAWAY LANDS** in a period known as the Age of Discovery, from the 1400s to the 1600s.



Marine sextant

Sextants (meaning sixths) use **MIRRORS** to measure the angle of the Sun or the North Star in relation to the horizon at particular times of day. Like the astrolabe, this allows sailors to work out their ***north–south position***. The first one was made by English astronomer **John Bird** in 1757. They are still used today—if onboard computers crash, mariners can **FIND THEIR WAY** with a sextant.

Did you know?
Marshall Islanders memorized stick charts, made from coconut fronds, to map ocean swells and navigate the Pacific by canoe.

The arc of a sextant is one sixth of a circle (60 degrees).

Satellite navigation

Today, there are ***networks of satellites*** in space that allow users to pinpoint their position almost anywhere on earth. A receiver compares **TIME SIGNALS** from four or more satellites. To determine its exact location, the receiver calculates the **distance to each satellite**. Today, most sailors rely on satellites to safely navigate through the world's waters—and many cars and cell phones have satellite receivers, too.